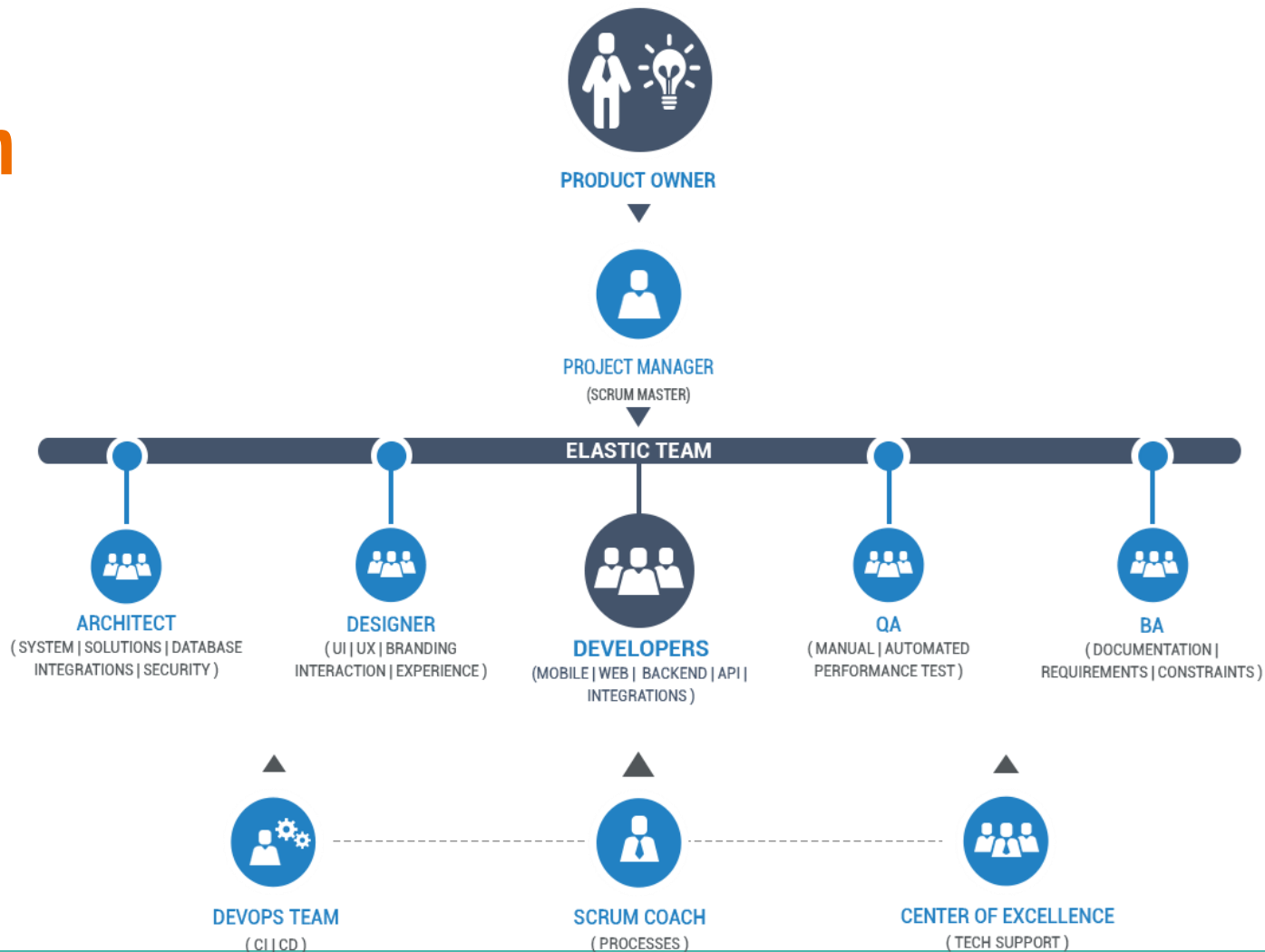

Software Engineering

Start-up!

Team



Why Software Engineering Project?!

1. Engineering - Software.
2. Current market status.
3. Importance of project?
 - a. Learn about Real-life situations.
 - b. Learn with continuous support.
 - c. A step on your career or an alternative career.
 - d. Each moment spent in the project can be beneficial in practice.
 - e. A good entity on your CV.
 - f. Try something new.
 - g. Grades aren't the objective.

Communication Skills



Time Management



Problem-Solving Skills



Teamwork and Collaboration



Objectives [Team Work]

- Large Team.
- Separation of responsibilities and smooth integration.
- Divide system into multiple modules.

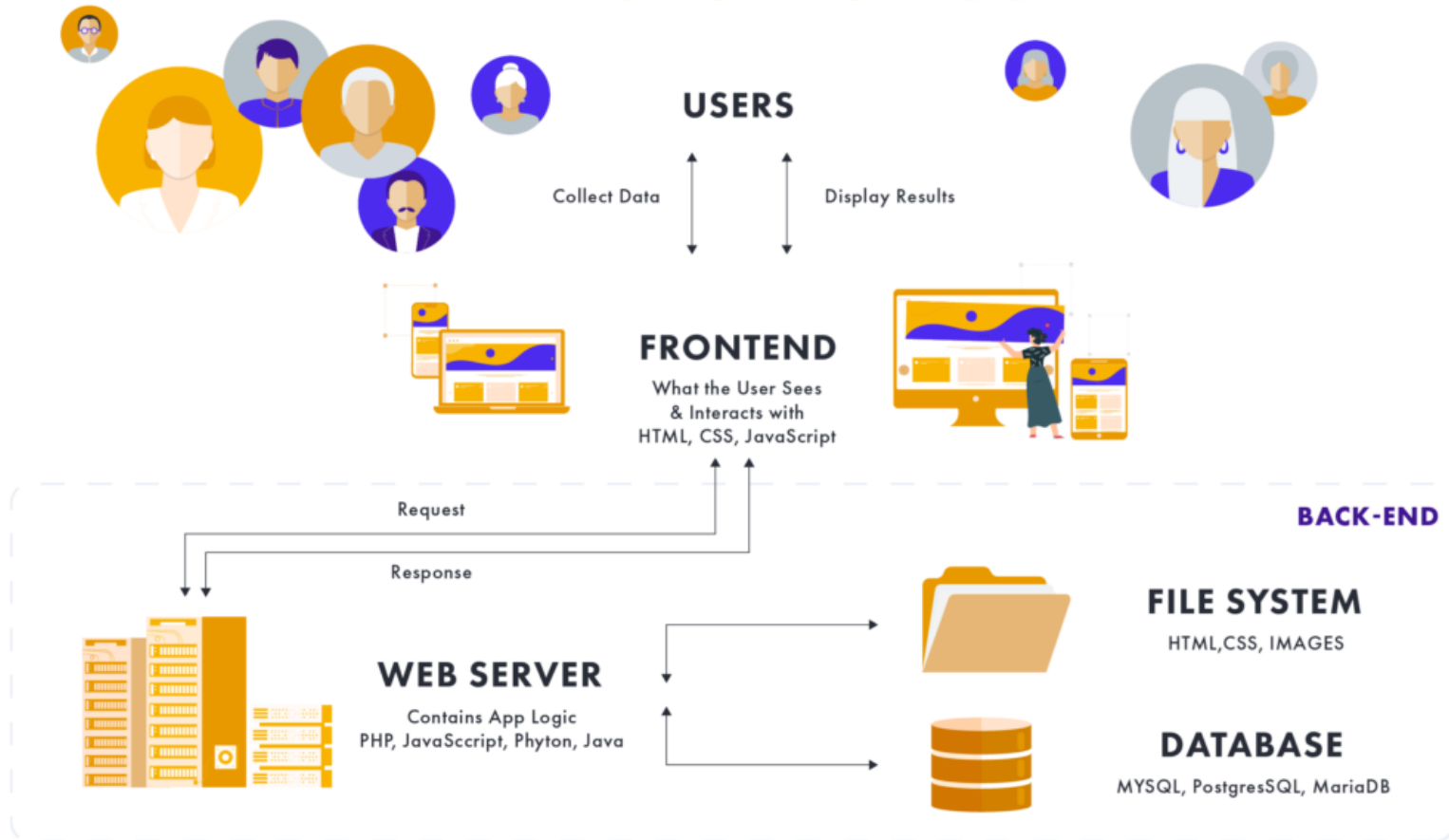
Teams

- Team Leader, is part of the Testing/DevOps
- Back-end : 4
- Front-end : 5/6
- DevOps : 1
- Mobile : 5
- Testing : 2

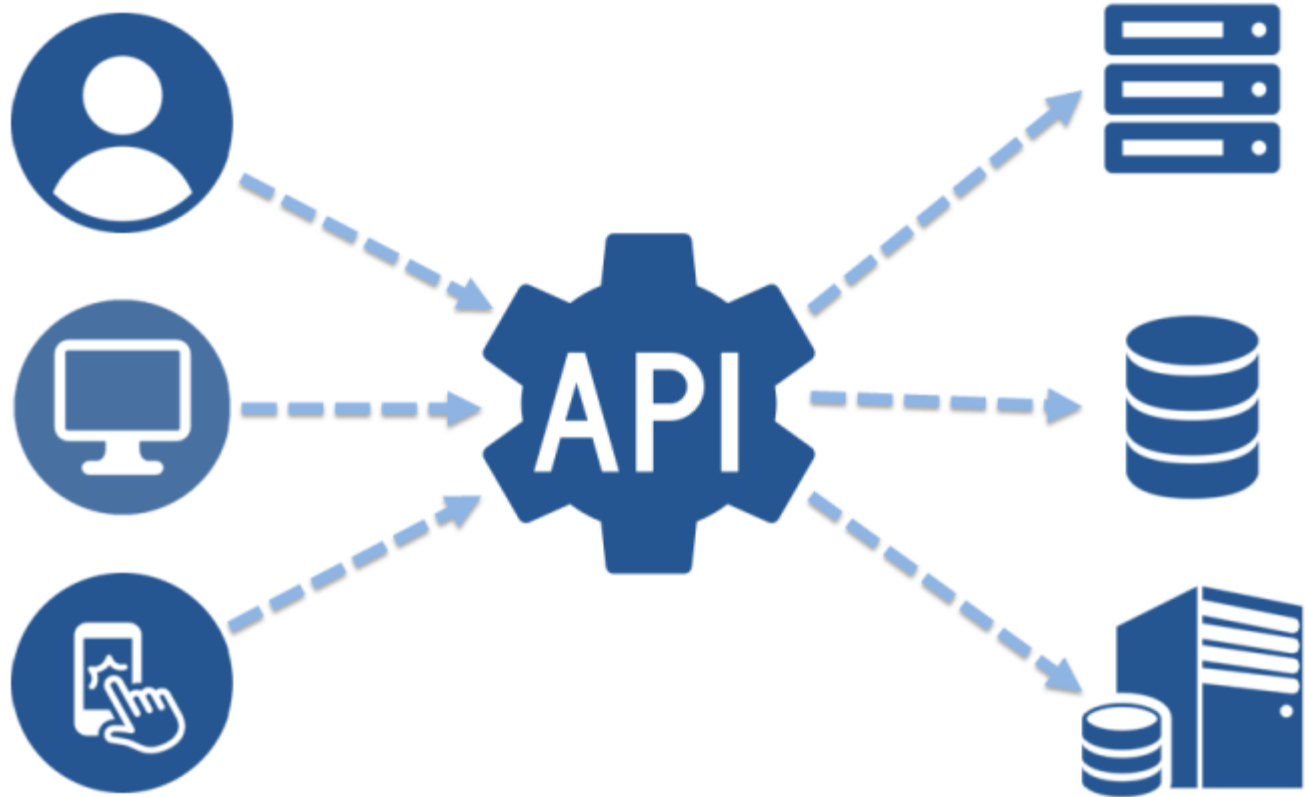
Team Leader

- Manage duties and tasks.
- Deadlines Determination and announcements.
- Synchronization & Communication with other sub-team leaders.
- Formal Communication.
- Main responsible.
- Feedback.

WEB APPLICATION ARCHITECTURE

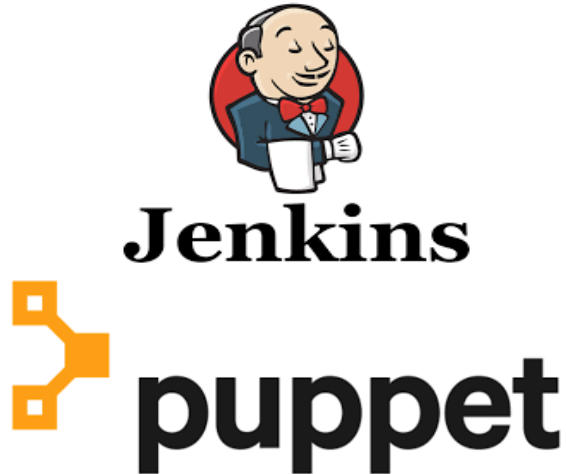


Backend Frontend API



DevOps Team

- Server Administration. (installation, update, solve problems and etc.)
- Monitoring & automation.
- Very quick deployment.



Testing Team

The Developer (Not the testing team)



Unit tests

Testing Team



E2E tests

Testing Team

- Test the product end-to-end [E2E].
 - Scripting.
 - Steps Recording [Not Accepted in Project]
- Stress Test.

Back-end Team

- DB.
- Handle requests.
- Authentication & Authorization.

Front-end & Android & iOS &

- Main tasks: GUI + Send requests $\Rightarrow$ Update GUI accordingly



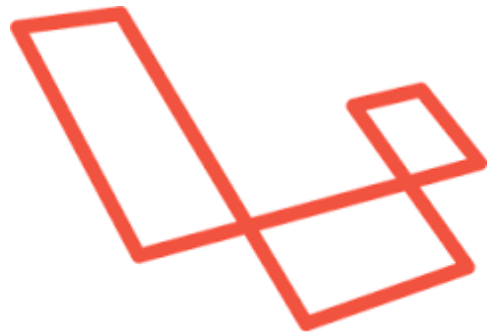
Back-end Team [DB Engines]



Back-end [Language & Frameworks]



Symfony



laravel



Back-end Team [Language & Frameworks]



Flask
django



Sprint Backlog

Forecast

To-Do

In-Progress

Done

Fix My Profile

5

aliquip

ipsum

duis

sit

ipsum

Filter Service
Tickets

8

dolor

ipsum

culpa

vale

culpa

aliquip

Quick Tips

3

ipsum

sit

duis

duis

Documentation

- Functional Documentation [BE, FE, Android, ...]
- API Documentation [BE]

Functional Doc.

Class HeroService

Hierarchy

- HeroService

Index

Constructors

- [+ constructor](#)

Properties

- | | | |
|---------------------------|--------------------------|----------------------------------|
| headers | http | translateService |
| heroesUrl | snackBar | translations |

Methods

- | | | |
|--------------------------------------|--------------------------------|--------------------------------|
| + checkIfUserCanVote | + getAllHeroes | + like |
| + createHero | + getHeroById | + showSnackBar |
| + deleteHeroById | + handleError | |

Globals

"heroes/shared/hero.service"

HeroService

- [+ constructor](#)
- [headers](#)
- [heroesUrl](#)
- [http](#)
- [snackBar](#)
- [translateService](#)
- [translations](#)
- [+ checkIfUserCanVote](#)
- [+ createHero](#)
- [+ deleteHeroById](#)
- [+ getAllHeroes](#)
- [+ getHeroById](#)
- [+ handleError](#)
- [+ like](#)
- [+ showSnackBar](#)

API Doc

JSONDoc

<http://beta.cytomine.be/restApiDoc/api#>

[Get documentation](#)

API INFO

Base path: <http://localhost:8080>

Version: 0.1

APIS

- [abstract image services](#)
- [acl services](#)
- [algo annotation services](#)
- [annotation filter services](#)
- [annotation index service](#)
- [annotation term service](#)
- [attached services](#)
- [description services](#)
- [discipline services](#)
- [generic annotation services](#)
- [group services](#)
- [image filter project services](#)
- [image filter services](#)
- [image group services](#)
- [image instance services](#)
- [image sequence services](#)
- [job parameter services](#)
- [job services](#)
- [job template annotation services](#)
- [job template services](#)
- [nested image services](#)

USER SERVICES

Methods for managing a user role

[/api/user/{id}.json](#)

GET

[/api/project/{id}/user.json](#)

GET

[/api/project/{id}/admin.json](#)

GET

Path

[/api/project/{id}/admin.json](#)

Description

Get all project admin

Method

GET

Produces

[application/json](#)

Path parameters

[id](#)

Required: true

Type: long

Description: The project id

Query parameters

[max](#)

Required: false

Type: int

Description: Pagination: Number of record per page (default 0 = no pagination)

[offset](#)

Required: false

Type: int

How to start?

- Form the teams.
 - Note: Sub-teams (not the whole team): Same grade (vertical & horizontal grading).
- Choose tech.
- Unify Tasks.

Phases

- Phase 0:
 - Used Tools & language. And why?
 - Team members and Positions.
 - Preparing repo & Task Management Tool.
- Phase 1:
 - Code Style for each team.
 - Task division (Use task management tool if appropriate) .
 - System Design [With the help of open APIs].

Phases

- Phase 2:
 - 20% Of **EVERY POINT**.
 - Progress Report (Individual + Team-based).
- Phase 3:
 - 50%
- Phase 4:
 - 100% + additions!

Recommendations

- <https://education.github.com/pack>.
- **Start learning early. Really, really early!**
- Divide learning between sub-team members.
- Use support at tutorials. It's part of the schedule.
- Single Duty "Be specific".
- Open Source Projects.
- Use the tools!



Google Summer of Code

Tips

- Transparency.
- Deep Understanding.
- Be Flexible:
 - No need to know everything before starting.
 - No need to know how to do every bit before starting.
 - Do the basic, Enhance, Iterate